

Maths behind Naive Bayes Algorithm

$$X = \{x_1, x_2, x_3 \dots x_n\} \text{ Features}$$

$$C_k = \{c_1, c_2, \dots c_k\} \text{ Target}$$

$$P(C_k | X) = \frac{P(X | C_k) P(C_k)}{P(X)}$$

$\Rightarrow$ This will be

common in each term

$$= P(X | C_k) P(C_k)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \therefore$$

$$P(A|B) P(B) = P(A \cap B)$$

chain Rule of conditional Probability

$$P(C_k | X) = P(X \cap C_k)$$

$$= P(\overset{A}{x_1}, \overset{\cap}{x_2, x_3 \dots x_n}, \overset{B}{C_k})$$

$$= P(X_1 | X_2, X_3 \dots X_n, C_k)$$

$$= \underbrace{P(X_1 | X_2, X_3 \dots X_n, C_k)}_a P(X_2, X_3 \dots X_n)$$

$$P(C_k | x) = a \times P(X_2, X_3 \dots X_n, C_k)$$

$$P(\overset{A}{\boxed{X_2}}, \overset{B}{\boxed{X_3 \dots X_n, C_k}})$$

$$P(C_k | x) = a \times \underbrace{P(X_2 | X_3 \dots X_n, C_k)}_b P(X_3, X_4 \dots X_n, C_k)$$

$$= ab P(X_3, X_4 \dots X_n, C_k)$$

$$= P(X_n | C_k) P(C_k)$$

$$P(X_{n-1} | X_n, C_k) P(X_n | C_k) P(C_k)$$

$$\underbrace{\hspace{15em}}_n$$

$$P(C_k | X) = P(C_k) \prod_{i=1}^n P(X_i | C_k)$$

$\prod$ = Product

$$P(C_k | X) = \frac{1}{Z} P(C_k) \prod_{i=1}^n P(X_i | C_k)$$

$$\hat{y} = \underset{k \in \{1, 2, 3 \dots k\}}{\operatorname{argmax}} P(C_k) \prod_{i=1}^n P(X_i | C_k)$$

Maximum a Posteriori Rule (MAP)